

BODEN, SwedenWMO: **021710**Lat: **65.815N**Lon: **21.640E**Elev: **82**StdP: **14.65**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-15.1	-27.2	1.2	-21.1	-20.5	1.8	-15.1	16.3	32.6	13.9	31.9	2.1	280	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	79.1	65.0	75.5	62.8	72.0	60.9	67.1	75.4	64.8	72.5	62.7	69.6	6.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	89.6	70.9	61.7	82.5	68.6	59.5	76.3	66.0	31.6	75.2	29.8	72.2	28.3	69.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	12.1	10.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-25.6	69.9	5.1	3.0	-29.3	72.0	-32.3	73.8	-35.1	75.5	-38.8	77.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	11.6	14.8	24.4	35.0	46.6	55.6	61.4	57.2	48.5	36.4	24.4	16.5
	DBStd	19.37	13.63	14.58	9.85	6.44	7.19	5.97	5.42	5.45	6.15	7.94	10.69	13.19
	HDD50	5917	1191	985	792	449	153	14	0	8	99	422	767	1037
	HDD65	10564	1656	1405	1257	899	570	289	140	249	494	886	1217	1502
	CDD50	869	0	0	0	0	48	181	354	230	55	1	0	0
	CDD65	39	0	0	0	0	0	5	28	6	0	0	0	0
	CDH74	460	0	0	0	0	27	64	312	57	0	0	0	0
	CDH80	66	0	0	0	0	2	9	51	4	0	0	0	0
Wind	WSAvg	4.6	3.8	4.3	5.2	4.7	5.5	5.4	4.7	4.4	4.5	4.4	3.9	4.1
Precipitation	PrecAvg	23.2	1.8	1.3	1.1	1.2	1.5	2.4	2.8	2.8	2.3	2.0	2.1	1.7
	PrecMax	34.2	4.3	2.3	2.4	3.0	4.6	4.5	5.3	6.1	7.6	5.3	3.9	4.8
	PrecMin	15.7	0.1	0.1	0.4	0.4	0.3	0.3	0.6	0.7	0.4	0.4	0.6	0.3
	PrecStd	4.8	0.9	0.6	0.6	0.6	0.9	1.0	1.2	1.4	1.5	1.0	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	46.1	48.7	63.1	77.1	80.7	84.6	80.2	68.4	56.5	48.1	40.9
		MCWB	40.3	40.5	41.1	49.5	58.1	65.4	68.0	68.1	59.5	49.1	42.9	37.9
	2%	DB	38.5	42.1	44.9	54.7	72.0	74.7	80.7	75.0	64.0	52.6	44.8	37.5
		MCWB	35.3	37.2	38.7	44.9	57.2	60.8	65.7	63.7	56.2	47.4	42.5	35.2
	5%	DB	33.9	37.9	42.2	50.9	65.9	71.3	77.6	71.1	61.2	49.9	41.5	35.7
		MCWB	32.1	34.6	36.5	42.3	53.9	58.8	64.6	62.1	54.7	46.6	39.3	34.0
	10%	DB	30.8	33.6	38.9	47.2	60.6	68.0	73.9	67.7	58.6	47.4	38.3	33.4
		MCWB	29.5	31.7	33.9	40.0	49.7	56.9	62.9	60.1	54.0	44.5	36.8	32.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	41.1	41.8	49.8	60.1	66.7	70.6	69.6	60.3	50.5	44.3	38.5
		MCDB	43.6	45.8	47.6	61.7	73.6	79.0	80.9	77.3	64.9	53.2	46.1	40.8
	2%	WB	35.8	37.7	39.0	45.6	57.7	62.6	68.0	65.9	58.0	48.9	42.4	35.6
		MCDB	38.2	41.6	43.8	53.3	70.0	71.7	76.5	72.5	62.0	51.5	44.9	36.8
	5%	WB	32.5	34.6	36.8	42.8	54.4	60.3	66.1	63.5	56.3	46.8	39.5	34.1
		MCDB	33.9	37.5	41.7	49.5	64.6	68.5	74.2	69.0	59.4	49.5	41.4	35.5
	10%	WB	29.4	32.1	34.2	40.5	51.0	58.4	64.1	61.1	54.8	44.7	36.8	32.1
		MCDB	30.7	33.7	38.0	46.8	58.8	66.4	72.0	65.8	57.7	47.2	38.2	33.2
Mean Daily Temperature Range	5% DB	MDBR	13.6	15.2	16.9	16.9	18.2	16.5	17.2	16.4	14.8	11.2	9.8	12.3
		MCDBR	14.7	16.2	17.9	23.1	27.7	22.0	24.1	21.6	19.1	14.7	11.4	12.0
		MCWBR	13.1	13.0	12.8	14.1	14.6	10.4	11.6	12.0	12.2	10.8	9.6	10.9
	5% WB	MCDBR	14.1	14.8	17.8	22.1	26.4	19.5	20.5	19.3	16.0	12.4	11.4	11.8
		MCWBR	12.9	12.3	13.0	13.9	14.4	10.2	11.0	11.8	11.3	9.4	10.0	10.9
Clear-Sky Solar Irradiance	taub	N/A	0.268	0.300	0.332	0.341	0.345	0.359	0.344	0.314	0.292	0.338	0.351	
	taud	N/A	2.154	2.236	2.331	2.461	2.485	2.465	2.496	2.558	2.397	2.651	2.675	
	Ebn at Noon	N/A	190	243	264	274	275	267	260	245	186	83	28	
	Edh at Noon	N/A	17	26	30	30	30	30	26	19	14	6	3	
All-Sky Solar Radiation	RadAvg	30	214	702	1277	1568	1699	1579	1170	662	248	52	6	
	RadStd	3	33	58	73	156	135	149	107	94	43	8	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page